

(point C) of the PIC in auto position (Fig. 1).

The buck-boost principle

Voltage stabilisers buck (subtract) the mains voltage if it is higher than 220V and boost (add to) the mains voltage if it is lower than 220V. For this purpose, you need to produce a small voltage to do addition or subtraction. In Fig. 3, the mains voltage waveform is shown

in the top left corner, with two voltages of smaller amplitude (about 30V) shown below it. One of these two voltages is in the same phase as the mains voltage, while the other is out of phase. By adding any of the two voltages, you can boost or buck the mains voltage.

For this purpose, ordinary voltage stabilisers generate a small voltage using a transformer with one or more taps. They connect the small voltage

in series with the mains supply so as to add or subtract from it. A change-over relay is used to switch to buck/boost, while another relay selects between voltages from the two taps.

This method does not produce a smooth voltage change due to relay switching and the voltage from tap produces a fixed value (instead of a finely-variable voltage). In this project, the additional voltage of about 30V in phase or out of phase with the mains voltage is finely variable be-

cause of PWM. So it produces a smoothly varying output.

A typical PWM concept is shown in Fig. 4. The microcontroller produces pulse-widths, as required, for generating the voltage to be added or subtracted from the mains. The pulses from points A and B (refer Fig. 1) are fed to the transformer shown in Fig. 2. The secondary winding of this transformer gives the adding voltage. In this case, there is no relay switching; the buck or boost is done smoothly by changing the phase of the adding signal instantly. So it is a continuous voltage stabiliser. Depending on how much the input varies from 220V, pulse width is generated so as to adjust the output voltage by adding or subtracting from it. This is a feed-forward control.

Points marked 'COM' common points in Figs 1 and 2 are not the ground and should not be connected to the neutral line.

Take care while checking the buck-boost circuit, as all the points are 'hot' and will give electric shock if touched, and also when interconnecting the voltage stabiliser control and buck-boost circuits.

Pulse-drive circuit and the transformer. Fig. 2 shows the circuit to buck/boost the mains voltage using a buck-and-boost transformer. The iron-core transformer used here is the same as used in voltage stabilisers. There is no tap on the secondary winding and the primary winding is centre-tapped.

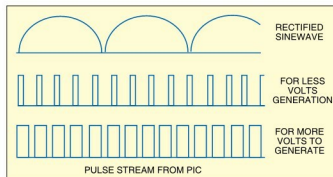
As with most transformers, the stampings used for this transformer are made of 4mm thick silicon steel. These are E-I type Stalloy/CRGO stampings. The size of the stampings depends on the rating. A toroidal-winding transformer gives better performance and is smaller in size.

Here, we have used a 250V-0-250V, 500mA primary to 50V, 5A secondary transformer. The windings' number of turns depends on the core size used.

Pulses from A and B of the voltage stabiliser controller circuit are fed to the gate pins of MOSFET power transistors (IRF840) via 10k series



(a) The pulse output of induced sine wave. This was taken with a 4MHz crystal to show the pulses clearly. The pulses will be finer in the actual 12MHz circuit



(b) Pulse-width variation with voltage

Fig. 4: A typical PWM concept

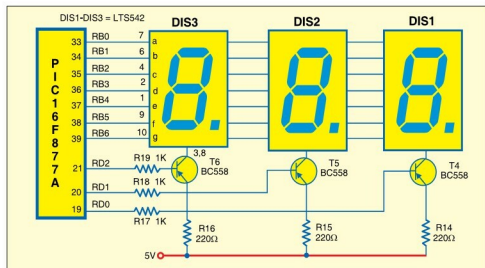


Fig. 5: Circuit for display section (optional)